

logs need to be shared in a common folder that can be accessed by the DCA server. The DCA will pick up the logs and send them to the PMA server for analysis. Initially, the PMA platform was designed to support Keysight ICTs since it had the default utility to communicate with the PMA server. Later, we expanded to include the whole manufacturing line by implementing the approach to read logs (generated by other machines) from the shared directory.

What are your thoughts on increasing automation of electronics manufacturing?

Not every aspect of a product necessitates automation. The degree of automation depends on the specific product. In the automotive sector, certain companies, particularly original equipment manufacturers (OEMs), insist on complete automation, driven by safety concerns and the desire to minimize human handling. Also, given the high cost of these products, such as high-end servers with a single printed circuit board assembly (PCBA) ranging from \$10,000 to \$20,000, OEMs advocate for full automation to reduce costs, lower the risk of accidental failures, and decrease the need for manpower. Even after 30 years, a blend of human and automated processes on surface mount technology (SMT) lines and automated optical inspection (AOI) is required. Despite advancements, complete eradication of human intervention from manufacturing seems unlikely. Some critical PCBA types may undergo full automation. While technically proficient individuals in India are prepared to support such automation, there will always be products that don't need full automation.

Which industry segments are driv-

ing maximum demand for your ICT solutions?

We see customers in the automotive, computing and netcomm business, along with aerospace, military, and medical segments, driving the demand. Since these are mission-critical applications, our PMA Analytics can immediately provide the history of the fault, reducing the time it would have otherwise taken if done manually.

Who are your target customers in India?

We are primarily focused on electronic manufacturing services (EMS) companies and original design manufacturers (ODM). Big companies like Google, Meta, Amazon, and Tesla are outsourcing their manufacturing to these companies. The bulk of our business comes from this set of manufacturers. In India, we are targeting local manufacturers. Regardless of size, we are committed to contributing to the success of everyone in the manufacturing industry.

Who is driving the demand in India—local manufacturers or foreign players setting up bases in India?

Right now, there are a lot of global companies coming in. The growth is there because they are bringing in the investment and have a lot of resources and money to put into operation. But I'm also seeing that local manufacturing companies have started to adapt and want to be global players in manufacturing. I have seen what happened in China before it became a global player in manufacturing, and I am witnessing the same scenario in India, with the government spending money on infrastructure, logistics, education, and resources for manufacturing.

How many team members does your manufacturing division have in India?

At Keysight, for electronic manufacturing test solutions, we have a global operation of 100+ resources, including local resources in India who have 10+ years of experience in ICT. We have also partnered with DJK. This partnership with DJK enables us to support most of our customers who are spread out in India.

In India, many electronics manufacturers are remotely placed. How do you ensure prompt support in such a vast geographic market?

We have designed our ICT system, 3070, to last and work 24x7 for many years. We have 20-year-old systems in manufacturing and operating, highlighting the reliability of the system. When it's down, we have a ready support network that can be activated immediately and respond to customers for assistance if they are under the support contract. If there is a deeper defect, we have a local support person who can fly to any location to help the customer resolve the system failure.

How do you see the growth of electronics manufacturing ecosystem in India? How does Keysight intend to contribute?

The Indian government has set huge targets to become a global manufacturing destination. Luring all the big companies into India is a big task, but we are witnessing it happen gradually. We have been associated with major global technological companies for over 40 years, including OEMs and EMS players. We intend to help them start off their manufacturing lines in India, duplicating what they have in other countries. At the same time, we are strengthening the capabilities of local manufacturers. For India to grow, they must grow the local manufacturing sector. **EFY**